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GEOGRAPHIC INFORMATION SYSTEMS IN POST-EARTHQUAKE TRANSPORTATION MANAGEMENT: A SYSTEMATIC REVIEW OF INTEGRATION STRATEGIES AND APPLICATIONS

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ABSTRACT

Managing transportation systems in the aftermath of earthquakes poses critical challenges that significantly impact emergency response operations and community recovery. Geographic Information Systems (GIS), which are tools for capturing, analyzing, and visualizing spatial data, have emerged as indispensable technologies for enhancing the efficiency and effectiveness of post-earthquake mobility management. This review paper explores the integration of GIS in addressing transportation challenges following seismic events. It examines the impacts of earthquakes on transportation infrastructure, evaluates the benefits and limitations of GIS applications, and identifies best practices and strategies for improving transportation system resilience. Additionally, the study highlights emerging trends and innovations in GIS technologies that have the potential to transform post-earthquake transportation management. Through a comprehensive analysis of case studies and scholarly literature, this paper aims to provide policymakers, practitioners, and researchers with valuable insights to support informed decision-making and foster the effective use of GIS in post-disaster contexts.

Keywords: transportation, Geographical Information System (GIS), post-earthquake.

INTRODUCTION

Managing transportation systems after earthquakes presents significant challenges, particularly for emergency response and community recovery. Ensuring the resilience of transport infrastructure is critical to facilitate the movement of people, supplies, and aid [1,2]. Damage to roads and bridges caused by earthquakes can delay response efforts and disrupt logistics [1]. GIS provide spatial data analysis tools that support post-earthquake transport management through emergency planning and coordination [3,5]. When integrated with seismic risk models, GIS aids in identifying vulnerable areas and optimizing resource allocation [4]. Historically, earthquakes have severely damaged transportation networks, highlighting the necessity of effective recovery planning [6]. Reliable transportation is essential for evacuation, rescue, and timely delivery of aid to high-risk areas [6]. Additionally, the growing demand for faster, safer, and more efficient transportation systems is driven by economic growth and globalization [8]. Initially used for damage mapping, GIS now enables real-time monitoring by integrating data from sensors, remote sensing, and social media [7,9]. This integration allows decision-makers to assess road conditions and population movements, thereby optimizing transportation routes. GIS integration enhances the resilience and

effectiveness of transportation systems by providing critical data, facilitating coordination, and enabling real-time spatial analysis in post-disaster scenarios.

Understanding Post-Earthquake Transportation Challenges

Different earthquake types cause distinct damage to transportation infrastructures such as bridges, tunnels, railway embankments, and roads, often leading to traffic disruptions and compromised critical facilities due to ground collapses and cascading geohazards [10]–[13]. The severity and duration of impacts depend on earthquake magnitude and location, resulting in blocked highways, disrupted ports, reduced bridge capacity, and limited emergency access [14]. The resilience of essential infrastructure—including hospitals, power, transport, and telecommunications—determines the effectiveness of post-disaster response [15]–[18]. Improved post-earthquake cordon management is vital to expedite disaster response and recovery by controlling public access [19]. Advances in mobility management have been informed by understanding coseismic and post-seismic landslide activities, which are critical to maintaining safe transport networks in vulnerable regions [7], [20], [21]. Medical logistics, including transport of the injured, is a key component of disaster response [21]. Case studies from Wellington and Christchurch demonstrate challenges in cordon implementation over time and space, emphasizing the need for enhanced operational protocols to ensure efficient access control and resource flow during post-earthquake recovery [19].

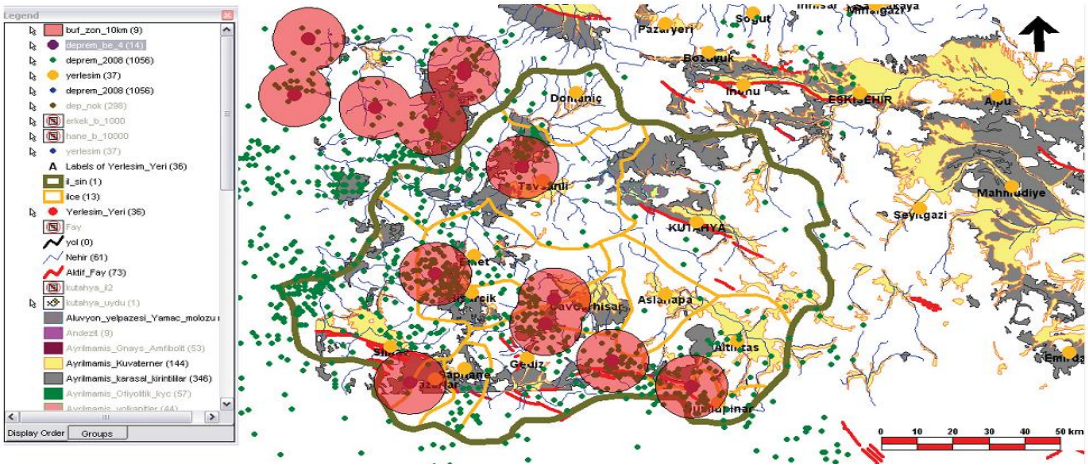


Figure 1. Kutahya Earthquake Buffer Zones and Road Map [68]

In the previous study about Kutahya, GIS technology and earthquake information system application were examined. As can be understood from the map created as a result of the Buffer Analysis, it is seen that the regions with earthquake magnitude 4 and above intersect with each other. With this visualization, it can be said that the earthquake risk areas of Kütahya province are high in the south and southwestern parts. It is seen in Figure 1 that the main roads are included in these buffer zones. With GIS maps, alternatives for accessing the main transportation roads from settlements in disasters can be found. This shows the importance of GIS in disaster management.

Role of GIS in Transportation Management

GIS are tools designed for data storage, digitization, querying, and reporting [22]. In disaster management, particularly post-earthquake transportation challenges, GIS is vital for analyzing and visualizing geographic data by integrating multiple components and features. A core function of GIS is the collection and storage of spatially referenced data, including buildings, roads, infrastructure, topography, and population density, often gathered via GPS-based platforms [23]. This geographic information is stored in databases for rapid retrieval and visualization. GIS enables the aggregation and presentation of complex data—such as population distribution, disaster risk, and transportation conditions—on unified maps [16]. GIS facilitates data acquisition from diverse sources, including maps and field surveys, and supports analysis of spatial relationships among variables [16]. It enhances preparedness and situational awareness for stakeholders involved in post-earthquake transport management by integrating demographic, hazard, and infrastructure data [4]. Visualization of spatial data via maps and visual aids further aids decision-making [24]. As an information system, GIS manages extensive spatial data and aids authorities in risk mitigation and identifying high seismic risk areas through spatial distribution analysis of seismic hazards [25], [26]. For example, GIS combined with multi-agent systems has been employed to simulate post-earthquake Urban Search

and Rescue (USAR) operations dynamically [5]. In transportation management, GIS supports data-driven strategies and evidence-based decisions, optimizing resource allocation, logistics, and infrastructure repair by identifying priority areas and distributing resources accordingly [27]. GIS also assists in locating optimal sites for emergency response centers, distribution hubs, and temporary shelters based on proximity and existing infrastructure [28].

Long-term, GIS contributes to improved decision-making by providing a comprehensive, geographically explicit overview of resources, infrastructure, and networks [19]. It enables evidence-based resilience planning and investment prioritization in transport infrastructure. For instance, [3] developed a systematic method using GIS to assess the seismic risk of urban buildings, aiding pre-earthquake scenario planning [16]. Understanding the vulnerability of transport assets is essential in post-earthquake traffic management, as it guides efficient allocation of repair and reconstruction resources.

GIS for Post-Earthquake Transportation Management

After an earthquake, Geographic Information Systems (GIS)—computer systems designed for spatial data collection, storage, querying, analysis, and visualization [23]—enable rapid mapping and assessment of transportation network damage. By integrating diverse data sources such as high-resolution satellite imagery, GIS facilitates comprehensive damage evaluations of roads, bridges, tunnels, and other infrastructure [29]. This spatial visualization allows stakeholders to prioritize repair and reconstruction efforts effectively by identifying critical areas requiring immediate attention [30]. GIS supports decision-makers by providing spatial context for damage severity and distribution, enabling efficient resource allocation. By overlaying data on key facilities—hospitals, emergency centers, transport hubs—GIS aids in prioritizing infrastructure essential for emergency response and logistics [31]. However, data accuracy and resolution variability pose challenges to reliable analysis, necessitating the use of up-to-date and precise datasets. In post-earthquake recovery, GIS enhances logistics planning by assessing highway degradation and bridge functionality [32], supporting emergency departments in deploying personnel and equipment through integration of real-time staffing and resource data [33]. Coupled with advanced modeling and simulation, GIS allows decision-makers to evaluate resource allocation scenarios, identify network vulnerabilities, and develop contingency plans to expedite restoration of transport services and supply chains [34], [35]. Incorporating stakeholder preferences into GIS-based frameworks improves resilience planning by addressing region-specific risks and vulnerabilities [36]. The use of artificial neural networks for landslide susceptibility mapping further underscores the importance of accurate geospatial data in mitigating geohazard impacts on transportation [37].

Real-time sensor data and satellite imagery provide critical, current information on seismic activity, structural damage, and environmental hazards, offering a synoptic overview of affected areas [38], [39]. Social media feeds augment situational awareness with on-the-ground reports and assistance requests. Remote sensing tools complement these datasets by quantifying damage levels to infrastructure [40]. The integration of real-time field data into GIS significantly enhances the resilience and operational efficiency of transportation networks in seismically vulnerable regions.

GIS Integration Challenges in Post-Earthquake

GIS use in post-earthquake mobility management faces challenges including the need for accurate, current data on soil, rainfall, elevation, and slope, which are difficult to obtain after disasters [41]. Resource limitations further hinder timely recovery and data-driven decisions [42]. Data collection and interpretation are complicated by potential errors and interoperability issues among systems, impeding stakeholder collaboration [43]–[46]. Communication and coordination delays are common in resource-constrained, dynamic environments, emphasizing the need for effective stakeholder management [47], [48]. Technical barriers such as software integration and system compatibility require multidisciplinary efforts to develop interoperable GIS platforms for disaster response [49].

The success of GIS-based transport management depends on real-time access to reliable spatial data, enabling rapid damage assessment and prioritized recovery [50]. Centralized data repositories with standardized formats are critical for improving data sharing and collaboration among diverse disaster response actors [51].

Post-Earthquake Transportation Management: Best GIS Integration Practices

Effective coordination among stakeholders is essential for seamless GIS integration in post-earthquake transportation management, improving response efficiency and decision-making through accurate spatial data and remote sensing [52]. Multi-agency collaboration among government, emergency services, transportation authorities, and communities enhances resource sharing and coordinated strategies [53]. Predictive modeling using historical earthquake, geological, and infrastructure data supports risk assessment and resilience planning by forecasting damage and evaluating infrastructure vulnerability via scenario-based simulations [2], [54]. Hazard mapping and risk assessment identify high-risk areas and potential impacts, guiding prioritized resource allocation and mitigation

within GIS frameworks [55], [56]. Vulnerability analysis of physical, social, and economic factors further informs long-term resilience strategies by highlighting transport network weaknesses in seismically active regions [56].

Future Directions in Post-Earthquake Transportation Management: GIS Integration

The integration of LiDAR technology with UAV-based remote sensing marks a significant advancement in post-earthquake transportation management. LiDAR captures precise 3D coordinate data—including topography, vegetation, and infrastructure—enabling the creation of high-resolution point clouds and digital elevation models essential for assessing seismic impacts [57], [58]. It also provides critical measurements such as vegetation height and sub-canopy snow levels relevant to traffic management [59]. Machine learning and deep learning techniques, including convolutional neural networks (CNNs), efficiently process large UAV datasets to classify aerial imagery, rapidly identifying damaged infrastructure and transit routes [60]. These technologies facilitate detailed 3D modeling that supports resilient transportation planning by offering comprehensive insights into terrain and infrastructure [61]. AI enhances susceptibility modeling by addressing nonlinear complexities, helping locate earthquake-prone areas and vulnerable transport assets; integration with GIS further aids resource allocation and maintenance planning [62], [63]. However, challenges remain in integrating heterogeneous datasets—demographic, transportation, seismic history—due to disparities in availability, quality, and format, affecting predictive model reliability [64]. Collaborative GIS platforms promote stakeholder engagement and support the development of digital resilience plans informed by spatial data and predictive modeling [65]. Multidisciplinary approaches, involving environmental science and geography, help overcome knowledge gaps and biases by expanding research beyond protected or tourist zones [66]. More broadly, remote sensing through ground and aerial sensors facilitates diverse Earth observations critical for disaster management [67].,

RESULTS

The thorough examination of transportation management following earthquakes with the integration of GIS has made it clear that the efficient application of GIS has greatly improved the evaluation, tracking, and restoration of transportation infrastructure following seismic occurrences. The major conclusions highlight the crucial role that GIS plays in enabling timely and focused response efforts to handle the complex problems brought on by post-earthquake transportation disruptions. GIS provides real-time spatial data analysis, decision support, and cooperative coordination among stakeholders.

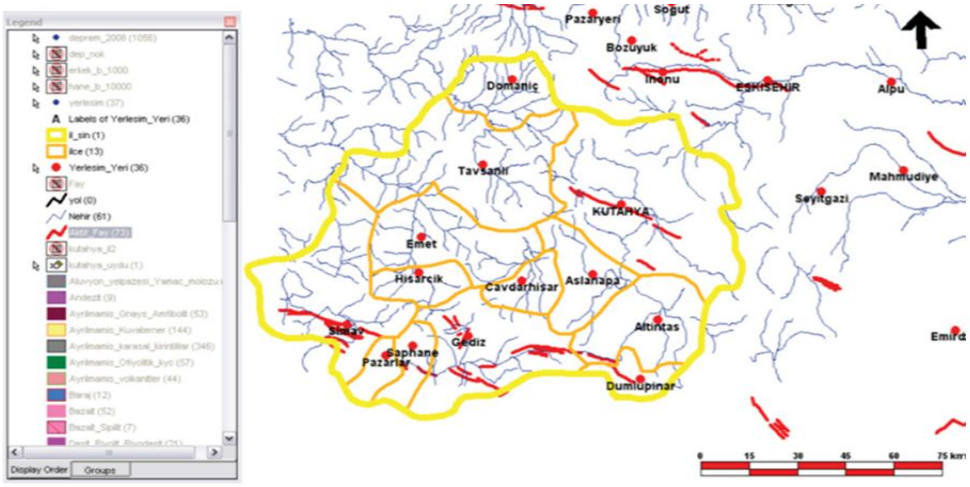


Figure 2. Kutahya Active Earthquake Fault Line Maps [68].

The red polyline layer in Figure 2 represents active fault lines concentrated around Kütahya city center, Simav, Gediz, Dumlupınar, and the southwestern parts of Şaphane districts, with some faults overlapping major road alignments [68]. This spatial overlap highlights areas where transportation infrastructure is at high risk of earthquake damage. The recent frequency of earthquakes and aftershocks in Simav is linked to these active faults, prompting transportation planners to consider potential disruptions and develop alternative road scenarios. [68].

CONCLUSIONS

- GIS integration has proven to be of utmost importance in supporting the resilience and efficacy of transportation management following earthquakes. It has allowed stakeholders to prioritize infrastructure rehabilitation efforts, create proactive mitigation strategies, and allocate resources optimally.
- Decision-makers may ensure that communities in earthquake-prone areas always have access to, are connected to, and are safe from transportation networks by utilizing GIS technologies to create a more resilient and adaptive transportation system that can withstand and recover from the negative effects of seismic events.
- The development of standardized data protocols, improved data accessibility, and interdisciplinary collaboration among stakeholders are critical priorities for policymakers, practitioners, and researchers to strengthen GIS integration for improved post-earthquake transportation management.
- To encourage a more sustainable and integrated approach to post-earthquake transportation management, it is also advised that capacity-building programs be put into place, sophisticated GIS tools and technologies be used, and strong governance frameworks be established. Establishing a robust and adaptable transportation infrastructure system that can successfully endure and bounce back from seismic occurrences requires proactive involvement from stakeholders in knowledge-sharing, technology transfer, and policy coordination.

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